

Validity Analysis: MILU

Target context: Mymensingh Environmental Scientist — Cross-Regional South Asian Agricultural Knowledge

Overall risk: **HIGH**

Deployment Context

Use case: Task: Environment science and geograpy-specific agricultural knowledge understanding

LLMs evaluated: Generic frontier model vs small llms as well as region specific llms

Domain: Environmental science

Setting: Environment science research utility assessment using llm and local-context

Target population: An environment scientist from Mymensingh, Bangladesh wants to get knowledge about different south asian region specific agricultural practice. So having different region and language specific agricultural and demographic context understanding (e.g. telegu spoken area, hydrabad India and agricultural practice/problems vs farmers practice/problems from Mymensingh, Bangladesh~bengali spoken in Mymensingh accented dialect.)

Dimension Scores

Dimension	Score	Rating	Confidence
Input Ontology $\triangle$	2	Significant gaps	high
Input Content $\triangle$	1	Serious concern	high
Input Form $\checkmark$	3	Moderate gaps	high
Output Ontology	2	Significant gaps	high
Output Content $\triangle$	2	Significant gaps	high
Output Form $\checkmark$	3	Moderate gaps	high
Average	2.2		

$\triangle$ = highest concern $\checkmark$ = strongest dimension

Dimension Key

Abbr.	Dimension	Definition
IO	Input Ontology	Whether the benchmark's test case categories cover the query types expected in deployment.
IC	Input Content	Whether individual datapoint content is culturally and contextually appropriate for the target region.
IF	Input Form	Whether the input signal encoding (text, audio, image parameters) matches deployment conditions.
OO	Output Ontology	Whether the benchmark's output categories and scoring criteria reflect regionally valid decision boundaries.
OC	Output Content	Whether ground-truth labels align with the judgments of regional stakeholders.
OF	Output Form	Whether the expected output modality matches regional deployment needs and accessibility.

Overall Summary

MILU is fundamentally misaligned with the Mymensingh environmental-scientist deployment context. The benchmark is by design India-centric across all six validity dimensions: its 41-subject taxonomy lacks Bangladesh-specific agro-ecological

subdomains; its content originates exclusively from Indian competitive exams with no Bangladeshi farming, dialect, or trans-border hydrology representation; its Bengali corpus reflects West Bengal exam register and is heavily machine-translated (100% of the sampled validation Bengali split); its annotation pool excluded Bangladeshi subject-matter experts; and its single-correct-answer MCQ format cannot accommodate the cross-border knowledge pluralism that trans-border water-policy questions demand. Dataset analysis empirically confirms these structural gaps: Bangladesh appears in 215 sampled examples only as a wrong-answer distractor, Agriculture-labeled questions concern Indian schemes and corporate trivia rather than agronomy, and Telangana/AP agro-ecological content is absent. MILU's strengths — broad Indic-script coverage, consistent MCQ structure, broad domain diagnostic breadth — make it useful as a generic Indic-LLM screening tool, but it provides essentially no discriminative signal for Bangladeshi agricultural and environmental science knowledge.

Practical Guidance

What This Benchmark Measures

MILU measures Indian competitive-exam general knowledge across 11 Indic languages and 8 broad domains. For the Mymensingh deployment, it can usefully measure (a) whether candidate LLMs render Bengali and Telugu scripts correctly (input_form strength) and (b) whether they have basic STEM and general-knowledge capability across Indic languages (output_form strength enables cross-model comparison). It cannot measure agricultural science depth, Bangladeshi agro-ecological knowledge, trans-border policy reasoning, or Bangladeshi Bengali register competence — the core constructs the deployment cares about.

Construct Depth

Construct depth is shallow for the target deployment. Even within the relevant Environmental Sciences domain, sampled content is geography trivia and Indian-policy-scheme recall, not agronomic, soil-science, or wetland-ecology reasoning (input_ontology score 2, input_content score 1). The Agriculture subject label exists but maps to institutional and corporate trivia. MILU provides surface-level breadth without the technical depth a professional environmental scientist would need to assess LLM reliability for their query workload.

What Else You Need

Significant supplementation is required: (a) a Bangladesh-specific agricultural knowledge benchmark covering haor/beel/char-land systems and boro/aman cycles (none currently exists per GAP-1, GAP-8); (b) BEnQA [WEB-6] as a Bangladeshi Bengali register baseline to disentangle language-competence from cultural-knowledge gaps; (c) a trans-border policy benchmark with pluralistic ground-truth structure to address output_ontology and output_content gaps; (d) Telangana/AP sub-regional agro-ecological items if cross-reference zones are core to the use case; and (e) generation-mode and CoT evaluation to address [Q85] log-likelihood limitations for reasoning-heavy queries.

ID	Evidence
Q85	"Finally, our evaluation primarily relies on the log-likelihood approach, which may yield different results compared to other established methods, such as generation-based evaluation and chain-of-thought (CoT) prompting." (p.9)
WEB-6	BEnQA exists as a separate Bangladeshi Bengali science benchmark, but is not integrated into MILU (https://arxiv.org/abs/2403.10900)

Input Ontology (IO)

Score: 2/5 — Significant gaps | Confidence: high

Benchmark: MILU | Context: Mymensingh Environmental Scientist — Cross-Regional South Asian Agricultural Knowledge

Input Ontology definition: Whether the benchmark's test case categories cover the query types expected in deployment.

Each section below is followed by an evidence table. Three types of evidence are cited: **[QN]** — verbatim quotes extracted from the benchmark paper; **[WEB-N]** — web sources consulted during assessment; **[DN]** / **[DATASET-DN]** — sampled datapoints from the benchmark dataset, with an interpretation note.

Justification

MILU's 41-subject taxonomy includes an Environmental Sciences domain and an Agriculture subject, but both are surface-level and India-scoped by design [\[Q19, Q38\]](#). The taxonomy lacks any Bangladesh-specific agro-ecological subdomains — no haor/beel wetland category, no char-land farming, no boro/aman rice cycle subject, and no trans-border water management category. Dataset analysis confirms this empirically: across 215 sampled examples, Environmental Sciences is populated almost entirely with geography trivia [\[DATASET-D6, D9, D16, D18\]](#) rather than agro-ecological science, and the few Agriculture-labeled questions concern Indian schemes/institutions [\[DATASET-D10, D13, D14\]](#). The authors themselves note poor model performance in 'culturally relevant areas' [\[Q6, Q68\]](#) which the deployment population would consider their core expertise. For a HIGH-priority deployment requiring agro-ecological depth across Bangladeshi delta and Indian cross-reference zones, the taxonomy is materially underrepresentative.

ID	Evidence
Q6	"Domain-wise analysis indicates that models perform poorly in culturally relevant areas like Arts & Humanities and Law & Governance compared to general fields like STEM." (p.1)
Q19	"MILU is designed as a culturally relevant benchmark to assess general problem-solving abilities and India-specific knowledge." (p.3)
Q38	"Following the manual merging of related clusters, we determine 41 distinct subject names, which fall into eight main domains: Arts and Humanities, Social Sciences, Environmental Sciences, Law and Governance, Health and Medicine, Science, Engineering and Technology, and Business Studies." (p.4)
Q68	"Similar trends to those in Section (§5.2) are observed where the open models perform poorly in domains specific to Indian culture—such as Arts & Humanities, Social Sciences, and Law & Governance—but demonstrate higher performance in STEM fields." (p.7)
D6	A geographic region has the following distinct characteristics: 1. Warm and dry climate 2. Mild and wet winter 3. Evergreen oak trees — Mediterranean climate region identification — generic geography
D9	Which state became the first Indian state to launch pension for single women at Rs.1,000/- per month? ... Telangana — Telangana state welfare policy — classified under Geography, reflects generic Indian current affairs
D10	domain: Environmental Sciences, subject: Agriculture — Agriculture-labeled question about India's National Food Security Mission — only MILU "Agriculture" example in Bengali sample
D13	राष्ट्रीय मोहरी संशोधन केंद्र कुठे स्थिति आहे... सेवर — Location of National Mustard Research Centre in India — India-specific agricultural institution
D14	ନିମ୍ନଲିଖିତ ଯେଉଁ କମ୍ପାନୀ ଦ୍ରୁତ ଗତିରେ କୃଷି ଉପକରଣ ଉତ୍ପାଦନ କରନ୍ତି?... ଦ କିରଲୋସ୍କାର — Which Indian company produces agricultural equipment rapidly (Kirloskar) — India-specific corporate knowledge
D16	మధ్యప్రదేశ్‌లోని బన్మోర్ పారిశ్రామిక అభివృద్ధి కేంద్రం ఏ జిల్లాలో ఉంది?... మౌరసా — Location of Banmore Industrial Development Centre in Madhya Pradesh — Indian geography
D18	domain: Environmental Sciences, subject: Geography — Mediterranean climate question under Environmental Sciences — geographic reasoning, not agro-ecological depth

Strengths

- MILU's taxonomy does include an Environmental Sciences domain and an Agriculture subject label that appears across multiple Indic languages, providing at least minimal coverage of the relevant domain category [\[Q38, DATASET-D10, DATASET-D13\]](#)

- The 8-domain × 41-subject structure offers broad coverage breadth that enables comparative diagnosis of model capabilities across general knowledge areas, useful as a baseline screening tool [Q1, Q18]

ID	Evidence
Q1	"MILU spans 8 domains and 41 subjects across 11 Indic languages, reflecting both general and culturally specific knowledge." (p.1)
Q18	"This benchmark covers many domains, including Science, Social Sciences, Humanities, Arts, Business Studies, and Law, among others." (p.3)
Q38	"Following the manual merging of related clusters, we determine 41 distinct subject names, which fall into eight main domains: Arts and Humanities, Social Sciences, Environmental Sciences, Law and Governance, Health and Medicine, Science, Engineering and Technology, and Business Studies." (p.4)
D10	domain: Environmental Sciences, subject: Agriculture — Agriculture-labeled question about India's National Food Security Mission — only MILU "Agriculture" example in Bengali sample
D13	राष्ट्रीय मोहरी संशोधन केंद्र कुठे स्थिति आहे... सेवर — Location of National Mustard Research Centre in India — India-specific agricultural institution

Checklist

Checklist item definitions:

ID	Assessment Question
IO-1	Identify test case categories required for regional deployment
IO-2	Check if taxonomy omits regionally relevant categories
IO-3	Check if taxonomy includes irrelevant categories
IO-4	Document category gaps harming content validity

Checklist responses:

IO-1: Required categories for the deployment include: haor/beel wetland ecology, char-land floodplain agriculture, boro/aman/aus rice cultivation systems, trans-border river hydrology (GBM system), Telangana dry-land cropping (cotton/sorghum/pigeonpea on Deccan red/black soils), Andhra Pradesh coastal aquaculture (L. vannamei/P. monodon under CAA/MPEDA frameworks), and integrated crop-fish farming. None of these appear as named subjects in MILU's 41-subject taxonomy [Q38].

IO-2: Yes — the taxonomy omits all Bangladesh-specific agro-ecological categories. The deployment YAML confirms this is structurally absent (GAP-1 resolution: 'CONFIRMED STRUCTURALLY ABSENT' [WEB-17]), and dataset analysis confirms no haor/beel/char/boro/aman content appears in 215 sampled examples (CRITICAL Concern 1: DATASET-D10, D13, D14, D15).

IO-3: Many MILU subjects are India-centric civic/cultural categories (Indian constitutional law, Indian state polity, medieval Indian history) that introduce construct-irrelevant variance for an environmental-science deployment. Dataset evidence: DATASET-D2 (1991 Indian fiscal crisis), D5 (Indian Election Commissioner removal), D19 (Bahmani Kingdom), D27 (Radcliffe Line), D28 (44th Constitutional Amendment) — none relevant to agricultural/environmental science.

IO-4: The category gap encompasses (a) absent wetland/floodplain ecology subdomains, (b) absent trans-border hydrology subdomain, (c) absent technical agronomy depth (soil science, crop physiology, irrigation) — only policy/institutional trivia appears under 'Agriculture' (CRITICAL Concern 1, MAJOR Concern 5). This constitutes substantial construct underrepresentation for the deployment.

ID	Evidence
Q6	"Domain-wise analysis indicates that models perform poorly in culturally relevant areas like Arts & Humanities and Law & Governance compared to general fields like STEM." (p.1)
Q19	"MILU is designed as a culturally relevant benchmark to assess general problem-solving abilities and India-specific knowledge." (p.3)

Q38	"Following the manual merging of related clusters, we determine 41 distinct subject names, which fall into eight main domains: Arts and Humanities, Social Sciences, Environmental Sciences, Law and Governance, Health and Medicine, Science, Engineering and Technology, and Business Studies." (p.4)
D2	১৯৯১ সালে আর্থিক সংকটের কারণ হিসেবে বদেশি মুদ্রার রজিার্ড হ্রাসের তাৎক্ষণিক কারণ কী ছিল? — India's 1991 foreign reserve crisis — Indian economic history, irrelevant to Bangladesh agricultural deployment
D5	নর্বাচন কমিশনারকে অপসারণ করা যত্নে পার... প্রধান নর্বাচন কমিশনারের সুপারিশে রাষ্ট্রপতি — Removal of Indian Election Commissioner — Indian constitutional law only
D10	domain: Environmental Sciences, subject: Agriculture — Agriculture-labeled question about India's National Food Security Mission — only MILU "Agriculture" example in Bengali sample
D13	রাষ্ট্রীয় মোহরী সংশোধন কেন্দ্র কুঠে স্থতি আছে... সেবর — Location of National Mustard Research Centre in India — India-specific agricultural institution
D14	নর্দিল্লিগতি নর্দ্রুদ কুঠে ঠালিও দত্তগতরিত কষ্ট্র উওকরগা উত্ওদন করত?... ও করিল্লোয়কার — Which Indian company produces agricultural equipment rapidly (Kirloskar) — India-specific corporate knowledge
D15	রাষ্ট্রীয় খাদ সুরঁখািা মস্টিন ঈব্ ডমল ব্খিম জেননা ঐ... মর্টি এী সযিও তুঁ ব্খাল বরনা — National Food Security Mission in Punjabi — same cross-language India-scoped agriculture question
D19	বাহামানি সাম্রাজ্যের প্রথম রাজধানী কী ছিল?... গুলবার্গা — First capital of Bahmani Kingdom — medieval Indian history, no agriculture/environment relevance
D27	রেডক্লিফি রেখা নীচেনা পৈকী ক্খা ঈশ সাথে ভারতনী সর্ভট্টেনে অলগ করে ঔ?... পাকিস্তান — Radcliffe Line divides India from Pakistan — Indian partition geography
D28	রাষ্ট্রীয় আপাতকাল ঘোষতি করনে কে লঈ 'সশস্ত্র ব্দিরোহ' শব্দ সংবধিন মে কব জোড়া গয়া?... 44বে সংবধিন সংশোধন অধনিয়িম দ্বারা — Indian constitutional amendment for 'armed rebellion' clause — Indian law and governance
WEB-17	AI4Bharat/MILU GitHub: no Bangladeshi annotator participation (https://github.com/AI4Bharat/MILU)

Information Gaps

- Exact subject-level question counts in Environmental Sciences for Bengali and Telugu test split (Table 9 not directly accessible)
- Whether any Telugu test questions include Telangana/AP sub-regional agro-ecological content (sample of 25 validation examples cannot rule in/out)

Requires Expert Verification

- A Bangladeshi agronomist's assessment of which specific subdomains would be required at minimum for a baseline professional-knowledge benchmark

Remediation

Gap 1: 41-subject taxonomy lacks Bangladesh-specific agro-ecological subdomains (haor wetland ecology, char-land agriculture, boro/aman rice systems, trans-border hydrology)

Recommendation: Collaborate with BAU Mymensingh and BARI/BRRI to define a Bangladesh-extended subject taxonomy with named subdomains for floodplain ecology, wetland farming, and cross-border river systems; construct an extension benchmark covering those subdomains.

Input Content (IC)

Score: 1/5 — Serious concern | Confidence: high

Benchmark: MILU | Context: Mymensingh Environmental Scientist — Cross-Regional South Asian Agricultural Knowledge

Input Content definition: Whether individual datapoint content is culturally and contextually appropriate for the target region.

Each section below is followed by an evidence table. Three types of evidence are cited: [QN] — verbatim quotes extracted from the benchmark paper; [WEB-N] — web sources consulted during assessment; [DN] / [DATASET-DN] — sampled datapoints from the benchmark dataset, with an interpretation note.

Justification

Every datapoint in MILU originates from Indian competitive exams [Q9, Q11, Q20, Q24, Q93]. The deployment YAML and web search confirm structural absence of Bangladeshi content (GAP-1, GAP-6 [WEB-17, WEB-18]). Dataset analysis confirms this empirically: across 215 sampled examples, Bangladesh appears only as a wrong-answer distractor in two questions about Indian Pharmacopoeia [DATASET-D21, D25], and no Bangladeshi institution, ecological system, or trans-border policy appears anywhere in the sample (CRITICAL Concern 1). Bengali content is West Bengal exam-derived, with the entire validation Bengali sample being machine-translated from English (DATASET-D3, all 21 examples). Telugu content shows no Telangana/AP agro-ecological specificity (MAJOR Concern 6). For a HIGH-priority dimension targeting Bangladeshi farming context, Mymensingh dialect terminology, haor wetlands, and trans-border hydrology, the content is essentially zero — a fundamental misalignment.

ID	Evidence
Q9	"We designed MILU with an India-first perspective by collecting questions from various national, state, and regional exams." (p.2)
Q11	"We create this benchmark by collecting questions from over 1500 competitive exams from India." (p.2)
Q20	"These questions were sourced following an approach similar to AGIEVAL (Zhong et al., 2023), collecting the questions from various public exams taken by individuals intending to either pursue higher studies or seek career advancements, such as qualification tests and national and state-level civil services exams, among others." (p.3)
Q24	"Regional state exams are particularly valuable as they cover various state-level topics and emphasize the official language of each state." (p.3)
Q93	"We collected our questions from over 40 exam types ranging from various National and state level civil service examinations to examinations conducted by various government and private organizations." (p.17)
D3	**is_translated:** True — All 21 Bengali validation examples are machine-translated from English
D21	ଦିଗେଶ୍ୱର 2019 ରେ ଭାରତୀୟ ଫାର୍ମାକୋପୋଇଆକୁ ପ୍ରଥମେ କଣ ଦେଖିବା ଦିଶେ କରେ? ... ଆଫଗାନିସ୍ତାନ; option2: ବାଙ୍ଗଳାଦେଶ — Which country first reported Indian Pharmacopoeia (Afghanistan) — Bangladesh appears as wrong answer option
D25	డిసెంబర్ 2019లో భారతీయ ఫార్మాకోపీయా ను గుర్తించిన మొదటి దేశం ఏదీ? ... ఆఫ్ఘనిస్తాన్; option2: బంగ్లాదేశ్ — Same Indian Pharmacopoeia question in Telugu — Bangladesh again used only as distractor
WEB-17	AI4Bharat/MILU GitHub: no Bangladeshi annotator participation (https://github.com/AI4Bharat/MILU)
WEB-18	NAACL paper: annotation by AI4Bharat volunteers and Indian exam-portal experts only (https://aclanthology.org/2025.naacl-long.507/)

Strengths

- MILU does include English questions specifically chosen to address South Asian culture-specific content [Q26], offering some pan-South-Asian general knowledge value
- West Bengal-derived Bengali content shares deltaic ecology context with Mymensingh and may carry partial transferable agronomic relevance for the West Bengal cross-reference zone (though depth is unconfirmed)

ID	Evidence
Q26	"English questions are also included as these often address Indian culture-specific content, which is notably missing from existing popular benchmarks." (p.3)

Checklist

Checklist item definitions:

ID	Assessment Question
IC-1	Do inputs require region-specific cultural/geographic/dialectal knowledge?
IC-2	Does culturally sensitive content align with target culture?
IC-3	Flag inputs assuming Western-specific knowledge
IC-4	Regional annotator recruitment for culturally sensitive instances
IC-5	Document content issues harming content validity

Checklist responses:

IC-1: Yes — the deployment requires Bangladeshi agro-ecological knowledge (haor, beel, char-land, boro/aman cycles), Mymensingh dialect agricultural terminology, and cross-border GBM hydrology. None of this is present in the sampled MILU content (CRITICAL Concern 1) and the YAML confirms structural absence ([WEB-17] AI4Bharat/MILU GitHub).

IC-2: Cultural alignment is poor: MILU's Bengali corpus reflects West Bengal exam register conventions, not Bangladeshi register. The expert elicitation (A2) noted 'potentially slight implicit Indian Bengali stylistic bias' and confirmed dialect-specific Bangladeshi agricultural terminology is absent. Dataset analysis: 100% of sampled Bengali validation items are machine-translated [DATASET-D3], compounding register mismatch.

IC-3: Many inputs require Indian-specific knowledge that does not transfer: Indian constitutional procedure [DATASET-D5, D28], Indian state assembly elections [DATASET-D4], Indian medieval/colonial history [DATASET-D19, D38], Indian corporate brands [DATASET-D14, D33]. These contribute construct-irrelevant variance for an environmental-science evaluation.

IC-4: INSUFFICIENT DOCUMENTATION — no Bangladeshi annotators participated in MILU construction (GAP-6 resolution: 'CONFIRMED ABSENT' [WEB-17, WEB-18]). For the deployment, Bangladeshi agronomist review of any MILU-relevant subset would be required to flag culturally sensitive instances.

IC-5: Documented content issues: (a) zero Bangladesh-specific datapoints, (b) Bengali content sourced from West Bengal exams [Q24], (c) ~25% machine-translated overall [Q43] with the validation Bengali split observed at 100% translated [DATASET-D3], (d) Bangladesh appears only as distractor [DATASET-D21, D25]. These cumulatively violate content validity for the target deployment.

ID	Evidence
Q9	"We designed MILU with an India-first perspective by collecting questions from various national, state, and regional exams." (p.2)
Q11	"We create this benchmark by collecting questions from over 1500 competitive exams from India." (p.2)
Q24	"Regional state exams are particularly valuable as they cover various state-level topics and emphasize the official language of each state." (p.3)
Q43	"Table 2 shows the overall statistics of MILU. Of the total 79K questions, only 25% of questions are translated from English, with the remainder" (p.4)
D3	**is_translated:** True — All 21 Bengali validation examples are machine-translated from English
D4	ফেব্রুয়ারি- মার্চ ২০২২ এ অনুষ্ঠিত রাজ্য বর্ধমানসভা নির্বাচন, আম আদমি পার্টি (AAP)... পাঞ্জাব — Indian state assembly election — Indian polity, no Bangladesh relevance
D5	নির্বাচন কমিশনারকে অপসারণ করা যত্নে পার... প্রধান নির্বাচন কমিশনারের সুপারিশে রাষ্ট্রপতি — Removal of Indian Election Commissioner — Indian constitutional law only
D14	নিম্নলিখিত পদ্ধতি দ্রুত কৃষি যন্ত্রপাতি উৎপাদন করে... কেরল কোম্পানি — Which Indian company produces agricultural equipment rapidly (Kirlskar) — India-specific corporate knowledge
D19	বাহমানি সাম্রাজ্যের প্রথম রাজধানী কী ছিল?... গুলবার্গা — First capital of Bahmani Kingdom — medieval Indian history, no agriculture/environment relevance
D21	তথ্যেবর 2019 ৱে ভারতীয় ঔষধি তত্ত্বাবধায়ক দপ্তর দ্বারা প্রকাশিত কোন দেশের প্রথম ভারতীয় ঔষধি তত্ত্বাবধায়ক?... আফগানিস্তান; option2: বাংলাদেশ — Which country first reported Indian Pharmacopoeia (Afghanistan) — Bangladesh appears as wrong answer option
D25	ডিসেম্বর 2019-এ ভারতীয় ঔষধি তত্ত্বাবধায়ক দপ্তর দ্বারা প্রকাশিত কোন দেশের প্রথম ভারতীয় ঔষধি তত্ত্বাবধায়ক?... ভারত; option2: বাংলাদেশ — Same Indian Pharmacopoeia question in Telugu — Bangladesh again used only as distractor

D28	राष्ट्रीय आपातकाल घोषित करने के लिए 'सशस्त्र विद्रोह' शब्द संविधान में कब जोड़ा गया?... 44वें संविधान संशोधन अधिनियम द्वारा — Indian constitutional amendment for 'armed rebellion' clause — Indian law and governance
D33	T - Hub ଏକ ଡିଜିଟାଲ ରାଜ୍ୟ ସରକାରର ପ୍ରୟୋଗ... ଉତ୍ପାଦନା ନୁହେଁ ବରଂ ଉତ୍ପାଦନ ଦେବାକୁ ଏକ ପ୍ରୟୋଗ ଉପରେ କେନ୍ଦ୍ର — T-Hub Telangana state entrepreneurship tech incubator — Telangana-specific but non-agricultural
D38	प्रांतांमध्ये द्वैधशासन प्रणाली कोणत्या कायद्याने स्थापन केली?... भारत सरकार अधिनियम 1919 — Government of India Act 1919 establishing dyarchy — Indian colonial law
WEB-17	AI4Bharat/MILU GitHub: no Bangladeshi annotator participation (https://github.com/AI4Bharat/MILU)
WEB-18	NAACL paper: annotation by AI4Bharat volunteers and Indian exam-portal experts only (https://aclanthology.org/2025.naacl-long.507/)

Information Gaps

- Whether the test split (vs validation) contains a higher proportion of original (non-translated) Bengali questions (DATASET limitation 1)
- Quantitative subject-level question counts for Agriculture/Environmental Science in Bengali and Telugu test splits (GAP-4 not resolved [WEB-18])

Requires Expert Verification

- A Bangladeshi linguist's assessment of register/terminology drift in MILU's machine-translated Bengali items

Remediation

Gap 1: Zero Bangladeshi datapoints; Bengali corpus is West Bengal-derived; sampled Bengali validation is 100% machine-translated

Recommendation: Source new Bengali items from Bangladeshi institutional question banks (BAU, BARI, BRRI, BWDB) authored in Bangladeshi register; integrate BEnQA [WEB-6] as a complementary Bangladeshi Bengali baseline; explicitly flag and quarantine machine-translated items in any deployment screening report.

ID	Evidence
WEB-6	BEnQA exists as a separate Bangladeshi Bengali science benchmark, but is not integrated into MILU (https://arxiv.org/abs/2403.10900)

Input Form (IF)

Score: 3/5 — Moderate gaps | Confidence: high

Benchmark: MILU | Context: Mymensingh Environmental Scientist — Cross-Regional South Asian Agricultural Knowledge

Input Form definition: Whether the input signal encoding (text, audio, image parameters) matches deployment conditions.

Each section below is followed by an evidence table. Three types of evidence are cited: **[QN]** — verbatim quotes extracted from the benchmark paper; **[WEB-N]** — web sources consulted during assessment; **[DN]** / **[DATASET-DN]** — sampled datapoints from the benchmark dataset, with an interpretation note.

Justification

MILU's text-based MCQ format is technically compatible with the deployment's text-only modality, and Bengali and Telugu scripts are reliably rendered (DATASET strengths 1, 4). However, two concerns reduce alignment: (a) the exam-register MCQ phrasing differs from naturalistic professional-query style (elicitation A4), and (b) approximately 25% of questions are machine-translated [Q43], with the sampled Bengali validation split observed at 100% translated (DATASET-D3, CRITICAL Concern 2), introducing register artifacts and occasional pipeline quality issues including incomplete question stems (DATASET-D34, D35, D36 — MINOR Concern 8). For a MODERATE-priority dimension where script support is the floor requirement, MILU clears that bar but does not match the naturalistic register a Mymensingh domain expert would use.

ID	Evidence
Q43	"Table 2 shows the overall statistics of MILU. Of the total 79K questions, only 25% of questions are translated from English, with the remainder" (p.4)
D3	**is_translated:** True — All 21 Bengali validation examples are machine-translated from English
D34	"নচিৰে প্ৰশ্ন... সেই বকল্‌পট নিৰ্বাচন কৰুন যা প্ৰদত্ত শব্দৰে বপীৰীত অৰ্থ প্ৰকাশ কৰে" (question body missing the target word) — Antonym question — incomplete stem, illustrates quality issues in translated questions
D35	"নচিৰে বৰিতিটি দিউ যুক্তি এবং II সহ দেওয়া হয়ছে" (arguments not shown in question field) — Argument evaluation question — missing argument content, shows data quality gap in translated items
D36	The owner of the textile shop brought a... Calculator — Incomplete question stem — no meaningful question context present

Strengths

- Bengali and Telugu scripts are fully populated in 215 sampled examples (DATASET Strength 1: D3, D31), confirming script-level encoding compatibility for the deployment's primary languages
- Multiple cleaning layers were applied — INDICLID language ID, Unicode filtering, deduplication, manual spot-checks [Q30, Q31, Q32] — providing baseline data hygiene
- Consistent 4-option MCQ structure across all 11 languages enables reproducible cross-language and cross-model comparison (DATASET Strength 4: D6, D11)

ID	Evidence
Q30	"To remove incorrect language entries, we utilize a combination of INDICLID (Madhani et al., 2023) and Unicode-based filtering (Khan et al., 2024), ensuring that the questions are in the correct language." (p.4)
Q31	"To further refine the dataset, we remove any duplicate questions to retain only the unique ones." (p.4)
Q32	"As a final step, we manually verify a sample of questions from each language to ensure accuracy and correct any remaining errors." (p.4)
D3	**is_translated:** True — All 21 Bengali validation examples are machine-translated from English
D6	A geographic region has the following distinct characteristics: 1. Warm and dry climate 2. Mild and wet winter 3. Evergreen oak trees — Mediterranean climate region identification — generic geography
D11	ರಾಷ್ಟ್ರೀಯ ಆಹಾರ ಭದ್ರತೆ ಮಂಷನ್ ಒಂದು ಬೆಳೆ ಅಭಿವೃದ್ಧಿ ಯೋಜನೆ... ಮಣ್ಣಿನ ಆರೋಗ್ಯವನ್ನು ಪುನಃಸ್ಥಾಪಿಸಲು — National Food Security Mission crop scheme in Kannada — same Indian policy content replicated across languages

D31	"স্বথরি প্রবাহ ট্রান্সফরমার _____ ধরনরে। ... শলে" (is_translated: True) — Shell-type constant current transformer — engineering question, machine-translated from English
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Checklist

Checklist item definitions:

ID	Assessment Question
IF-1	Compare signal distributions between source and target
IF-2	Check if regional infrastructure supports data capture specs
IF-3	Identify domain-specific form differences
IF-4	Document form mismatches harming external validity

Checklist responses:

IF-1: The text-based signal modality matches deployment needs (both are text). However, register distribution differs: MILU is exam-register MCQ, deployment is naturalistic professional query (elicitation Q4/A4). Script encoding in Bengali and Telugu is verified functional [DATASET-D3, D31].

IF-2: Mymensingh has connectivity adequate for text-based LLM access — Bangladesh has ~47% internet penetration with ~100% 4G coverage as of 2025 [WEB-20, WEB-21], and Mymensingh hosts BAU and is a divisional capital [WEB-7]. Text MCQ infrastructure is supported.

IF-3: Domain-specific form differences: (a) machine-translated content carries register artifacts — the YAML notes 'translated questions likely carry Indian Bengali register conventions rather than Bangladeshi Bengali norms', and (b) some translated items have incomplete stems [DATASET-D34, D35] suggesting pipeline degradation. MCQ format also excludes reading-comprehension and image-based queries [Q29], though this is non-binding for the deployment.

IF-4: Form mismatches are moderate: register artifacts from machine translation [Q43, Q84] (DATASET-D3 — 100% translation in Bengali validation), occasional truncated question stems [DATASET-D34, D35, D36], and exam-register vs. naturalistic-query phrasing divergence. None block use, but they reduce ecological validity.

ID	Evidence
Q4	"Open multilingual models outperform language-specific fine-tuned models, which perform only slightly better than random baselines." (p.1)
Q29	"During the collection process, we exclude any reading-comprehension-style questions, images-based questions, and those with more than four answer options to ensure uniformity and consistency." (p.4)
Q43	"Table 2 shows the overall statistics of MILU. Of the total 79K questions, only 25% of questions are translated from English, with the remainder" (p.4)
Q84	"Third, the scarcity of questions necessitated translating a portion of the dataset." (p.9)
D3	**is_translated:** True — All 21 Bengali validation examples are machine-translated from English
D31	"স্বথরি প্রবাহ ট্রান্সফরমার _____ ধরনরে। ... শলে" (is_translated: True) — Shell-type constant current transformer — engineering question, machine-translated from English
D34	"নচিরে প্রশ্ন... সেই বকল্‌পটিনিবিচান কব্বুন যা প্রদত্ত শব্দরে বপিরীত অর্থ প্রকাশ করে" (question body missing the target word) — Antonym question — incomplete stem, illustrates quality issues in translated questions
D35	"নচিরে বব্বিতটিদুটযুক্তি এবং II সহ দেওয়া হয়ছে" (arguments not shown in question field) — Argument evaluation question — missing argument content, shows data quality gap in translated items
D36	The owner of the textile shop brought a... Calculator — Incomplete question stem — no meaningful question context present
WEB-7	BAU as professional institutional context — text infrastructure supported (https://bau.edu.bd/)
WEB-20	Bangladesh internet/mobile coverage adequate for text modality (https://datareportal.com/reports/digital-2024-bangladesh)
WEB-21	Mobile-dominant connectivity in Bangladesh (https://en.wikipedia.org/wiki/Internet_in_Bangladesh)

Information Gaps

- Whether the test split (vs validation) has a lower machine-translation rate for Bengali
- Mymensingh-specific connectivity statistics (national aggregates only [WEB-20])

Requires Expert Verification

- A Bangladeshi Bengali linguist's review of GPT-4O-translated MILU items for register naturalness vs. Indian Bengali bias

Remediation

Gap 1: Exam-register MCQ phrasing diverges from naturalistic professional-query register; some translated items have truncated stems

Recommendation: Augment evaluation with naturalistic-register Bangladeshi Bengali agricultural query items elicited from BAU/BARI scientists; filter out items with incomplete stems [DATASET-D34, D35, D36] before reporting scores.

ID	Evidence
D34	"নচিৰে প্ৰশ্ন... সেই বকিল্পট নিৰ্বাচন কৰুন যা প্ৰদত্ত শব্দৰে বপীৰীত অৰ্থ প্ৰকাশ কৰে" (question body missing the target word) — Antonym question — incomplete stem, illustrates quality issues in translated questions
D35	"নচিৰে ববিতটি দুটি যুক্তি এবং II সহ দেওয়া হয়েছে" (arguments not shown in question field) — Argument evaluation question — missing argument content, shows data quality gap in translated items
D36	The owner of the textile shop brought a... Calculator — Incomplete question stem — no meaningful question context present

Output Ontology (OO)

Score: 2/5 — Significant gaps | Confidence: high

Benchmark: MILU | Context: Mymensingh Environmental Scientist — Cross-Regional South Asian Agricultural Knowledge

Output Ontology definition: Whether the benchmark's output categories and scoring criteria reflect regionally valid decision boundaries.

Each section below is followed by an evidence table. Three types of evidence are cited: **[QN]** — verbatim quotes extracted from the benchmark paper; **[WEB-N]** — web sources consulted during assessment; **[DN]** / **[DATASET-DN]** — sampled datapoints from the benchmark dataset, with an interpretation note.

Justification

MILU's output ontology is single-correct-answer MCQ over 41 India-derived subject labels and 8 domains [Q35, Q36, Q37, Q38]. The structural single-correct-label format cannot accommodate cross-border knowledge pluralism on contested topics like trans-border water policy — the deployment YAML identifies this as a HIGH-priority concern (transboundary_policy_context, GAP-3 [WEB-11, WEB-14]). Even if such questions existed, the format would force a single nationally-partial answer key (MAJOR Concern 7: DATASET-D5, D28). The 41-subject labels themselves were derived by clustering ~20K Indian-exam tags [Q35, Q36] and validated against Indian standards. Domain-wise model performance is poor in culturally relevant areas [Q16, Q68], and the authors note a need for 'more inclusive data distribution' [Q70]. For a MODERATE-priority dimension, the structural rigidity plus India-derived label space creates meaningful misalignment, though the basic MCQ correctness structure is functional for non-contested factual content.

ID	Evidence
Q16	"Our domain-wise analysis reveals that models perform poorly in culturally relevant areas, such as Arts & Humanities and Social Sciences, compared to more general fields like STEM." (p.2)
Q35	"Finally, in total, we get around 20K tags. However, these tags are highly fine-grained, often having a heavy overlap." (p.4)
Q36	"To organize them, we embed the tags using the NV-EMBED-V2 (Lee et al., 2024) model and apply K-means clustering to group tags into 50 clusters." (p.4)
Q37	"We manually review these clusters and assign appropriate subject labels." (p.4)
Q38	"Following the manual merging of related clusters, we determine 41 distinct subject names, which fall into eight main domains: Arts and Humanities, Social Sciences, Environmental Sciences, Law and Governance, Health and Medicine, Science, Engineering and Technology, and Business Studies." (p.4)
Q68	"Similar trends to those in Section (§5.2) are observed where the open models perform poorly in domains specific to Indian culture—such as Arts & Humanities, Social Sciences, and Law & Governance—but demonstrate higher performance in STEM fields." (p.7)
Q70	"Bridging this gap requires a more inclusive data distribution that ensures equitable representation of all cultures and languages." (p.7)
D5	নর্বিচান কমিশনারকে অপসারণ করা যত্নে পার... প্রধান নর্বিচান কমিশনারের সুপারিশে রাষ্ট্রপতি — Removal of Indian Election Commissioner — Indian constitutional law only
D28	राष्ट्रीय आपातकाल घोषित करने के लिए 'सशस्त्र विद्रोह' शब्द संविधान में कब जोड़ा गया?... 44वें संविधान संशोधन अधिनियम द्वारा — Indian constitutional amendment for 'armed rebellion' clause — Indian law and governance
WEB-11	Ganges treaty in renegotiation 2024–2026 — Indian and Bangladeshi positions diverge (https://thediplomat.com/2026/04/why-the-india-bangladesh-ganges-treaty-renewal-must-deliver-real-security/)
WEB-14	No formal Teesta treaty; positions categorically differ (https://en.wikipedia.org/wiki/Teesta_Water_Dispute)

Strengths

- The 41-subject taxonomy was derived through systematic clustering and manual review of ~20K fine-grained tags [Q35, Q36, Q37], reflecting principled construction rather than ad-hoc category assignment
- Domain-wise reporting allows users to disaggregate model performance by domain, enabling targeted diagnosis of weak areas like Arts & Humanities and Law & Governance [Q16, Q68]

- Cross-language structural consistency in label space supports comparative evaluation (DATASET Strength 4: D11)

ID	Evidence
Q16	"Our domain-wise analysis reveals that models perform poorly in culturally relevant areas, such as Arts & Humanities and Social Sciences, compared to more general fields like STEM." (p.2)
Q35	"Finally, in total, we get around 20K tags. However, these tags are highly fine-grained, often having a heavy overlap." (p.4)
Q36	"To organize them, we embed the tags using the NV-EMBED-V2 (Lee et al., 2024) model and apply K-means clustering to group tags into 50 clusters." (p.4)
Q37	"We manually review these clusters and assign appropriate subject labels." (p.4)
Q68	"Similar trends to those in Section (§5.2) are observed where the open models perform poorly in domains specific to Indian culture—such as Arts & Humanities, Social Sciences, and Law & Governance—but demonstrate higher performance in STEM fields." (p.7)
D11	ರಾಷ್ಟ್ರೀಯ ಆಹಾರ ಭದ್ರತಾ ಮಿಷನ್ ಒಂದು ಬೆಳೆ ಅಭಿವೃದ್ಧಿ ಯೋಜನೆ... ಮಣ್ಣಿನ ಆರೋಗ್ಯವನ್ನು ಪುನಃಸ್ಥಾಪಿಸಲು — National Food Security Mission crop scheme in Kannada — same Indian policy content replicated across languages

Checklist

Checklist item definitions:

ID	Assessment Question
OO-1	Review output label categories for regional relevance
OO-2	Identify missing regionally specific categories
OO-3	Flag categories encoding non-regional values
OO-4	Consider stakeholder-driven taxonomy redesign
OO-5	Document taxonomy issues harming structural/content validity

Checklist responses:

OO-1: Output labels are India-validated and India-derived [Q35, Q38]. Regional relevance for Bangladesh is partial: factual STEM labels generally transfer, but governance/law/policy labels reflect Indian constitutional and administrative categories not applicable to Bangladesh [DATASET-D5, D8, D28].

OO-2: Missing categories: no Bangladesh-specific agro-ecological label space (haor, beel, char-land), no trans-border hydrology label, no cross-border policy category that could accommodate plural answer keys. Confirmed absent [WEB-17, WEB-18].

OO-3: Indian state polity labels (Maharashtra Legislative Committee, Indian Election Commissioner removal) encode Indian-specific governance assumptions [DATASET-D5, D8, D17]. For environmental policy, Indian exam answers would encode India's official water-sharing positions, which differ from Bangladeshi authoritative positions on the Ganges and Teesta [WEB-11, WEB-14].

OO-4: Stakeholder-driven taxonomy redesign would be required to add (a) wetland ecology subdomains, (b) cross-border policy category supporting plural ground truths, and (c) technical agronomy depth subdomains (soil science, crop physiology, irrigation engineering).

OO-5: Documented taxonomy issues: India-derived label space [Q35, Q38], single-correct-answer structural rigidity (MAJOR Concern 7), and absence of regional categories. These violate structural validity (taxonomy doesn't fit cross-border pluralism) and content validity (missing categories) for the deployment.

ID	Evidence
Q16	"Our domain-wise analysis reveals that models perform poorly in culturally relevant areas, such as Arts & Humanities and Social Sciences, compared to more general fields like STEM." (p.2)
Q35	"Finally, in total, we get around 20K tags. However, these tags are highly fine-grained, often having a heavy overlap." (p.4)

Q38	“Following the manual merging of related clusters, we determine 41 distinct subject names, which fall into eight main domains: Arts and Humanities, Social Sciences, Environmental Sciences, Law and Governance, Health and Medicine, Science, Engineering and Technology, and Business Studies.” (p.4)
Q70	“Bridging this gap requires a more inclusive data distribution that ensures equitable representation of all cultures and languages.” (p.7)
D5	নর্দ্বিবাচন কমশিনারকে অপসারণ করা যতে পার... প্রধান নর্দ্বিবাচন কমশিনারের সুপারশি রাষ্ট্রপতি — Removal of Indian Election Commissioner — Indian constitutional law only
D8	Which one of the following statements about Legislative Committee system of Maharashtra is not correct? — Maharashtra state governance — India-specific law, not relevant to Bangladesh
D17	కీరీండి హటిలో మహారాష్ట్ర శసనసభ కమిటీ వ్యయవనీధ గురించి ఏ వరకటన సరైనది కుదు? — Maharashtra Legislative Committee system in Telugu — state-specific Indian governance
D28	राष्ट्रीय आपातकाल घोषित करने के लिए 'सशस्त्र विद्रोह' शब्द संविधान में कब जोड़ा गया?... 44वें संविधान संशोधन अधिनियम द्वारा — Indian constitutional amendment for 'armed rebellion' clause — Indian law and governance
WEB-11	Ganges treaty in renegotiation 2024–2026 — Indian and Bangladeshi positions diverge (https://thedi diplomat.com/2026/04/why-the-india-bangladesh-ganges-treaty-renewal-must-deliver-real-security/)
WEB-14	No formal Teesta treaty; positions categorically differ (https://en.wikipedia.org/wiki/Teesta_Water_Dispute)
WEB-17	AI4Bharat/MILU GitHub: no Bangladeshi annotator participation (https://github.com/AI4Bharat/MILU)
WEB-18	NAACL paper: annotation by AI4Bharat volunteers and Indian exam-portal experts only (https://aclanthology.org/2025.naacl-long.507/)

Information Gaps

- Exact granularity of subject labels within Environmental Sciences (Agriculture, Geography, Earth Sciences, Environmental Science as separate sub-subjects?) — not directly observable beyond DATASET-D6, D7, D10

Requires Expert Verification

- A Bangladeshi policy expert's identification of which Indian exam answer keys would be considered authoritative vs. nationally partial for environmental governance questions

Remediation

Gap 1: Single-correct-answer MCQ structure cannot encode legitimate Indian–Bangladeshi pluralism on contested trans-border policy questions

Recommendation: For policy and trans-border items, adopt a multi-reference scoring schema where both Indian and Bangladeshi authoritative answers are accepted, or construct a separate pluralistic-ground-truth track using Bangladesh policy documents and the GBM bilateral context [WEB-11, WEB-14].

ID	Evidence
WEB-11	Ganges treaty in renegotiation 2024–2026 — Indian and Bangladeshi positions diverge (https://thedi diplomat.com/2026/04/why-the-india-bangladesh-ganges-treaty-renewal-must-deliver-real-security/)
WEB-14	No formal Teesta treaty; positions categorically differ (https://en.wikipedia.org/wiki/Teesta_Water_Dispute)

Output Content (OC)

Score: 2/5 — Significant gaps | Confidence: high

Benchmark: MILU | Context: Mymensingh Environmental Scientist — Cross-Regional South Asian Agricultural Knowledge

Output Content definition: Whether ground-truth labels align with the judgments of regional stakeholders.

Each section below is followed by an evidence table. Three types of evidence are cited: **[QN]** — verbatim quotes extracted from the benchmark paper; **[WEB-N]** — web sources consulted during assessment; **[DN]** / **[DATASET-DN]** — sampled datapoints from the benchmark dataset, with an interpretation note.

Justification

Ground-truth labels originate exclusively from Indian exam portals where 'subject experts ensure the accuracy of the answers' [\[Q22\]](#) and AI4Bharat volunteers conducted manual audits [\[Q87\]](#). No Bangladeshi annotators, agronomists, or environmental scientists are documented in the annotation pool (GAP-6 resolution: 'CONFIRMED ABSENT' [\[WEB-17, WEB-18\]](#)). Approximately 45% of questions arrived pre-tagged from source portals, with the remainder receiving GPT-4O-MINI-generated topic assignments [\[Q33, Q34\]](#). There is no documentation of inter-annotator agreement metrics or annotator demographics beyond AI4Bharat affiliation. The expert elicitation (A3) noted Indian exam answers are 'generally considered authoritative' for Bangladeshi environmental scientists, with notable exceptions around trans-border water agreements where Indian answers encode nationally partial perspectives — and the YAML documents that bilateral volatility (Ganges 2026 expiry, Teesta unresolved) makes this gap more acute [\[WEB-11, WEB-14\]](#). Dataset analysis confirms Bangladesh appears only as wrong-answer distractor [\[DATASET-D21, D25\]](#) — Bangladeshi perspectives are structurally excluded. For a HIGH-priority dimension, this is a substantial convergent-validity concern.

ID	Evidence
Q22	"These portals typically tag questions manually with topic names and language details, and subject experts ensure the accuracy of the answers." (p.3)
Q33	"Upon examination, we found that approximately 45% of questions were accurately labeled with a topic name, while the remaining questions lacked this information." (p.4)
Q34	"To address this issue, we first translate the untagged questions into English using INDICTRANS2 (Gala et al., 2023) and then prompt GPT-4O-MINI model to assign an appropriate topic name to the question." (p.4)
Q87	"We are also immensely grateful to the volunteers from the AI4Bharat team for their motivation and meticulous efforts in conducting manual audits." (p.9)
D21	ଡକ୍ଟ୍ରିନେର 2019 ରେ ଭାରତୀୟ ଫାର୍ମାକୋପିଆକୁ ପ୍ରଥମେ କଣ ଦଲେଇବା ଦେଖି କରେ?... ଆଫଗାନିସ୍ତାନ; option2: ବାଙ୍ଗଳାଦେଶ — Which country first reported Indian Pharmacopoeia (Afghanistan) — Bangladesh appears as wrong answer option
D25	డిసెంబర్ 2019లో భారతీయ ఫାର୍ମାకోପీయా ను గుర్తించిన మొదటి దేశం ఏదీ?... ఆఫ్ఘనిస్తాన్; option2: బంగ్లాదేశ్ — Same Indian Pharmacopoeia question in Telugu — Bangladesh again used only as distractor
WEB-11	Ganges treaty in renegotiation 2024–2026 — Indian and Bangladeshi positions diverge (https://thedi diplomat.com/2026/04/why-the-india-bangladesh-ganges-treaty-renewal-must-deliver-real-security/)
WEB-14	No formal Teesta treaty; positions categorically differ (https://en.wikipedia.org/wiki/Teesta_Water_Dispute)
WEB-17	AI4Bharat/MILU GitHub: no Bangladeshi annotator participation (https://github.com/AI4Bharat/MILU)
WEB-18	NAACL paper: annotation by AI4Bharat volunteers and Indian exam-portal experts only (https://aclanthology.org/2025.naacl-long.507/)

Strengths

- Indian exam-portal source provides documented subject-expert validation pathway for ~45% of pre-tagged questions [\[Q22, Q33\]](#), offering some baseline authority for non-contested STEM facts
- Manual cluster review for subject label assignment [\[Q37\]](#) and AI4Bharat volunteer audits [\[Q87\]](#) provide a layer of human oversight

- For non-contested scientific facts (basic chemistry, biology, physics), expert elicitation (A3) confirms Indian exam answers are generally accepted as authoritative by Bangladeshi environmental scientists

ID	Evidence
Q22	"These portals typically tag questions manually with topic names and language details, and subject experts ensure the accuracy of the answers." (p.3)
Q33	"Upon examination, we found that approximately 45% of questions were accurately labeled with a topic name, while the remaining questions lacked this information." (p.4)
Q37	"We manually review these clusters and assign appropriate subject labels." (p.4)
Q87	"We are also immensely grateful to the volunteers from the AI4Bharat team for their motivation and meticulous efforts in conducting manual audits." (p.9)

Checklist

Checklist item definitions:

ID	Assessment Question
OC-1	Do ground-truth labels reflect regional stakeholder perspectives?
OC-2	Assess potential annotator-regional population disagreement
OC-3	Review annotator demographics from documentation
OC-4	Consider label re-annotation by regional annotators
OC-5	Review aggregation methods for minority perspective erasure
OC-6	Document label issues harming convergent/external validity

Checklist responses:

OC-1: Ground truth labels reflect Indian exam-portal subject experts and AI4Bharat volunteers [[Q22](#), [Q87](#)]. For non-contested STEM facts, this is broadly acceptable to Bangladeshi scientists (elicitation A3). For trans-border policy, governance, and Bangladesh-specific agricultural practice, Indian answers encode partial perspectives [[WEB-11](#), [WEB-14](#)].

OC-2: Likely disagreement zones: (a) trans-border river-flow allocation answers, (b) agricultural-policy authority questions where Indian and Bangladeshi national positions differ, (c) Bangladesh-specific cropping-calendar questions (none currently exist in MILU). Elicitation A3 explicitly identified this. Currently, no such questions are present in sampled data, so the disagreement is latent rather than active.

OC-3: Annotator demographics documented are: Indian exam-portal subject experts [[Q22](#)], AI4Bharat team volunteers [[Q87](#)], and GPT-4O-MINI for topic-tag assignment [[Q34](#)]. No Bangladeshi annotators documented [[WEB-17](#), [WEB-18](#)]. No inter-annotator agreement metrics, no demographic breakdown beyond institutional affiliation. INSUFFICIENT DOCUMENTATION on annotator country/region distribution.

OC-4: Re-annotation by a Bangladeshi annotator pool would be required for any deployment-relevant subset, particularly for environmental policy and trans-border water questions. The deployment YAML notes BAU and BARI as candidate institutional sources [[WEB-7](#), [WEB-8](#)].

OC-5: Aggregation method is single-correct-answer MCQ, which by construction erases minority perspectives — a Bangladeshi alternative answer to a contested water-policy question would be marked wrong. Authors do not discuss aggregation alternatives or pluralistic labeling.

OC-6: Documented label issues: (a) no Bangladeshi annotator participation [[WEB-17](#)], (b) no IAA reported, (c) GPT-4O-MINI used for ~55% of topic tagging [[Q34](#)] without human verification documented per item, (d) Bangladesh appears only as distractor [[DATASET-D21](#), [D25](#)]. These violate convergent validity (labels don't correlate with regional perspectives where they would diverge) and external validity for the deployment.

ID	Evidence
Q22	"These portals typically tag questions manually with topic names and language details, and subject experts ensure the accuracy of the answers." (p.3)

Q34	"To address this issue, we first translate the untagged questions into English using INDICTRANS2 (Gala et al., 2023) and then prompt GPT-4O-MINI model to assign an appropriate topic name to the question." (p.4)
Q87	"We are also immensely grateful to the volunteers from the AI4Bharat team for their motivation and meticulous efforts in conducting manual audits." (p.9)
D21	ଡବିମେବର 2019 ରେ ଭାରତୀୟ ଫାର୍ମାକୋପୋଇଆ କେଉଁ ଦେଶ ଦେଖାଦେଇଛି?... ଅଫଗାନିସ୍ତାନ; option2: ବାଙ୍ଗଳାଦେଶ — Which country first reported Indian Pharmacopoeia (Afghanistan) — Bangladesh appears as wrong answer option
D25	డిసెంబర్ 2019లో భారతీୟ ఫାର୍ମାକୋପୋଇଆ ను గుర్తించిన మొదటి దేశం ఏదో?... ఆఫ్ఘనిస్థాన్; option2: బంగ్లాదేశ్ — Same Indian Pharmacopoeia question in Telugu — Bangladesh again used only as distractor
WEB-7	BAU as professional institutional context — text infrastructure supported (https://bau.edu.bd/)
WEB-8	https://en.wikipedia.org/wiki/Bangladesh_Agricultural_University
WEB-11	Ganges treaty in renegotiation 2024–2026 — Indian and Bangladeshi positions diverge (https://thediplomat.com/2026/04/why-the-india-bangladesh-ganges-treaty-renewal-must-deliver-real-security/)
WEB-14	No formal Teesta treaty; positions categorically differ (https://en.wikipedia.org/wiki/Teesta_Water_Dispute)
WEB-17	AI4Bharat/MILU GitHub: no Bangladeshi annotator participation (https://github.com/AI4Bharat/MILU)
WEB-18	NAACL paper: annotation by AI4Bharat volunteers and Indian exam-portal experts only (https://aclanthology.org/2025.naacl-long.507/)

Information Gaps

- No inter-annotator agreement metrics reported anywhere in MILU documentation
- No demographic breakdown of AI4Bharat volunteer auditors (gender, regional origin, professional discipline)
- Per-item verification rate for GPT-4O-MINI-generated topic tags [Q34]

Requires Expert Verification

- Bangladeshi environmental scientist re-annotation of any cross-border water or Bangladesh-specific agricultural items, were such items added
- Identification by a Bangladeshi policy expert of specific MILU items where Indian exam answers would be considered nationally partial vs. universally authoritative

Remediation

Gap 1: No Bangladeshi annotators; no inter-annotator agreement metrics; ~55% of topic tags assigned by GPT-4O-MINI without per-item human verification

Recommendation: Recruit a Bangladeshi annotator panel including BAU agronomists and an environmental-policy expert to re-annotate any items relevant to Bangladesh, trans-border water, or wetland systems; report inter-annotator agreement at minimum for the Bangladesh-relevant subset.

Output Form (OF)

Score: 3/5 — Moderate gaps | Confidence: high

Benchmark: MILU | Context: Mymensingh Environmental Scientist — Cross-Regional South Asian Agricultural Knowledge

Output Form definition: Whether the expected output modality matches regional deployment needs and accessibility.

Each section below is followed by an evidence table. Three types of evidence are cited: [QN] — verbatim quotes extracted from the benchmark paper; [WEB-N] — web sources consulted during assessment; [DN] / [DATASET-DN] — sampled datapoints from the benchmark dataset, with an interpretation note.

Justification

MILU evaluates via accuracy on MCQ tasks using log-likelihood scoring for non-API models [Q48–Q51] and structured-JSON generative scoring for API models [Q53]. Both deployment and benchmark are text-based, satisfying the basic modality match. The MCQ output format does not require speech, vision, or other modalities that would create infrastructural barriers in Mymensingh (high text-only connectivity [WEB-20]). Two limitations reduce the score: (a) the authors acknowledge log-likelihood may yield different results than generation-based or CoT methods [Q85], which matters for the deployment’s reasoning-heavy queries; and (b) MCQ output cannot capture the multi-step open-ended agricultural reasoning the deployment scenario requires (CRITICAL Concern 3, MAJOR Concern 7). For a LOWER-priority dimension where text modality is the main requirement, MILU is functionally adequate but suppresses richer output forms relevant to professional knowledge work.

ID	Evidence
Q48	“For non-API-based models, we use the LM-EVALUATION-HARNESS (Gao et al., 2024; Biderman et al., 2024) to ensure clean and reproducible evaluations.” (p.5)
Q49	“We use the log-likelihood method, where the probability of a given output string is computed by conditioning it on some provided input (Brown et al., 2020).” (p.5)
Q50	““Specifically, the log-likelihood of an answer (a) given the question (x), i.e., $\log P(a x)$ ” (p.5)
Q51	““For multiple choice questions, given k possible answer strings, we select the answer string (ai) with the highest conditional log probability, i.e., $\text{argmax}(\log P(a_i x))$ ” (p.5)
Q53	“We explicitly prompt these models to generate the correct response in a structured JSON format to simplify response parsing.” (p.5)
Q85	“Finally, our evaluation primarily relies on the log-likelihood approach, which may yield different results compared to other established methods, such as generation-based evaluation and chain-of-thought (CoT) prompting.” (p.9)
WEB-20	Bangladesh internet/mobile coverage adequate for text modality (https://datareportal.com/reports/digital-2024-bangladesh)

Strengths

- Text-based output modality is fully aligned with the deployment’s text-only LLM access pattern [Q48, Q53]
- Reproducible evaluation via LM-EVALUATION-HARNESS [Q48] and structured JSON parsing for API models [Q53] enables consistent benchmarking across the frontier and regional Indic LLMs the user deploys
- 0/1/5-shot evaluation across base and instruct variants [Q45, Q46] provides multiple data points on model behavior, including the documented warning that instruct models may degrade with few-shot examples [Q62, Q63] — useful diagnostic signal for the deployment
- Consistent MCQ structure supports cross-language and cross-model comparability (DATASET Strength 4)

ID	Evidence
Q45	“Both the base versions and instruction fine-tuned variants of these models, wherever applicable, are evaluated to measure the improvements gained from fine-tuning.” (p.5)
Q46	“All models, except for proprietary models and LLAMA-3.1-405B, are tested under 0-shot, 1-shot, and 5-shot setups.” (p.5)
Q48	“For non-API-based models, we use the LM-EVALUATION-HARNESS (Gao et al., 2024; Biderman et al., 2024) to ensure clean and reproducible evaluations.” (p.5)
Q53	“We explicitly prompt these models to generate the correct response in a structured JSON format to simplify response parsing.” (p.5)

Q62	"In contrast, Instruct models exhibit more varied behavior, where models either stagnate or even degrade in performance." (p.6)
Q63	"This also aligns with expectations, as Instruct models are specifically fine-tuned as conversation assistants and may not respond well to the few-shot in-context examples format." (p.6)

Checklist

Checklist item definitions:

ID	Assessment Question
OF-1	Does output modality match regional deployment needs?
OF-2	Assess text-to-speech availability for speech output
OF-3	Consider literacy rates and accessibility requirements
OF-4	Document form mismatches harming external validity

Checklist responses:

OF-1: Output modality (text MCQ answer-key selection) matches the deployment's text-based LLM access [Q48, Q53]. Modality match is good; format-richness match is partial — deployment expects open-ended reasoning, MCQ provides only label selection.

OF-2: Text-to-speech is not required by the deployment (text-based access only per YAML llm_access_modality). Not applicable.

OF-3: Target population is professional environmental scientists; literacy and accessibility are not constraints. Bangladesh national connectivity supports text modality [WEB-20, WEB-21]. Mymensingh hosts BAU, suggesting above-average professional digital infrastructure [WEB-7].

OF-4: Form mismatch is mild: log-likelihood vs. generative evaluation differences acknowledged by authors [Q85] mean MILU scores may not predict generation-mode deployment performance. MCQ format suppresses open-ended agricultural reasoning the deployment requires (MAJOR Concern 7). External validity for generation-mode deployment is partial.

ID	Evidence
Q48	"For non-API-based models, we use the LM-EVALUATION-HARNESS (Gao et al., 2024; Biderman et al., 2024) to ensure clean and reproducible evaluations." (p.5)
Q53	"We explicitly prompt these models to generate the correct response in a structured JSON format to simplify response parsing." (p.5)
Q85	"Finally, our evaluation primarily relies on the log-likelihood approach, which may yield different results compared to other established methods, such as generation-based evaluation and chain-of-thought (CoT) prompting." (p.9)
WEB-7	BAU as professional institutional context — text infrastructure supported (https://bau.edu.bd/)
WEB-20	Bangladesh internet/mobile coverage adequate for text modality (https://datareportal.com/reports/digital-2024-bangladesh)
WEB-21	Mobile-dominant connectivity in Bangladesh (https://en.wikipedia.org/wiki/Internet_in_Bangladesh)

Information Gaps

- Whether deployment workflow requires explicit chain-of-thought or generation-mode evaluation that would diverge from MILU's log-likelihood scoring [Q85]

Remediation

Gap 1: Log-likelihood MCQ scoring suppresses open-ended agricultural reasoning relevant to the deployment use case [Q85]

Recommendation: Add a generation-mode evaluation track with rubric-based scoring for a sub-sample of agriculturally relevant items, and include CoT prompts to align with the deployment's reasoning-heavy query workload.

ID	Evidence
Q85	“Finally, our evaluation primarily relies on the log-likelihood approach, which may yield different results compared to other established methods, such as generation-based evaluation and chain-of-thought (CoT) prompting.” (p.9)